Implementation Notes - Speech Understanding Assignment 2

Design Choice 1: Spectral Subtraction over Wiener Filtering

For classroom noise, spectral subtraction is computationally lighter and more interpretable than Wiener filtering. We use mag_clean = max(mag - noise factor * noise profile, 0.1*mag) with noise factor=2.0, tuned empirically for ~10dB SNR boost without over-suppression. Wiener filtering requires statistical noise model estimation; spectral subtraction works directly on magnitude spectra.

Design Choice 2: Frame-Level LSTM for LID (not frame-shift)

A naive frame-by-frame classifier is brittle to single-frame noise. Instead, we use a 2-layer LSTM over 20ms frames to capture temporal context. This naturally handles code-switching boundaries within ±100ms. Alternative: sliding window SVM required manual window tuning; LSTM learns optimal context automatically.

Design Choice 3: N-gram Logit Bias in Whisper Decoding

Direct vocabulary constraint is simpler than fine-tuning. We maintain a set of technical terms {"stochastic", "cepstrum", ...} and boost their logits during beam search. This avoids overfitting to small lecture-specific datasets. Fine-tuning would require 2+ hours of in-domain audio; bias requires only the vocabulary list.

Design Choice 4: Simple Linear DTW for Prosody Warping

Full DTW is $O(n^2)$ costly for 10-minute audio. We extract F0 and energy at 50ms hops (~12k frames), compute alignment, then resample target prosody. This is O(n log n) after sorting. Full path backtracking is skipped; we use only the final scaling factor, which is sufficient for perceptual prosody transfer.

Design Choice 5: LFCC over CQCC for Anti-Spoofing

LFCC (Linear Frequency Cepstral Coefficients) are computationally cheaper than CQCC and equally effective for synthetic speech detection. We extract 13-dim LFCC, use a shallow 3-layer MLP classifier, and compute EER on held-out bona-fide vs. synthesized pairs. CQCC adds Q-constant warping complexity without measurable gain on this task.

Design Choice 6: FGSM over PGD for Adversarial Testing

Single-step FGSM is fast (one gradient backward pass per epsilon) and sufficient to test LID robustness. Multi-step PGD would cost 10× more computation with diminishing returns for this framing. We add SNR constraint (target SNR > 40dB) to ensure perturbations remain inaudible.